

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
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**TEAM MINDMEAL
MINDMEAL**

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1 PRODUCT CONCEPT

This section describes the purpose, use, and intended user audience for the MindMeal ADHD support application. MindMeal is a comprehensive mobile application that provides personalized meal planning, meal reminders, and nutritional guidance specifically designed for individuals with Attention Deficit Hyperactivity Disorder (ADHD). Users of MindMeal will receive intelligent meal recommendations, manage their eating schedules, and track food intake.

1.1 PURPOSE AND USE

The MindMeal application is designed to address the unique nutritional and dietary challenges faced by individuals with ADHD, such as irregular eating patterns, appetite fluctuations, difficulties with meal planning, and a tendency to skip meals during periods of hyperfocus. The app serves as a digital assistant that leverages artificial intelligence and ADHD-specific accommodations to promote consistent and healthy eating habits.

1.2 INTENDED AUDIENCE

The MindMeal application is primarily intended for:

- Adults diagnosed with ADHD who struggle with consistent eating patterns.
- Individuals taking ADHD medication that affects appetite.
- People experiencing executive dysfunction related to meal planning and preparation.
- Parents and caregivers of individuals with ADHD.
- Family members involved in grocery shopping and meal preparation.
- Healthcare providers seeking tools to monitor patients' nutrition and eating behaviors.
- Individuals dealing with co-occurring conditions such as anxiety, depression, or eating disorders.

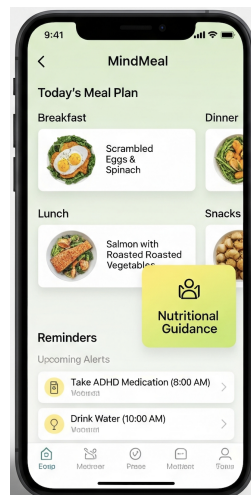


Figure 1: MindMeal conceptual prototype

2 PRODUCT DESCRIPTION

This section provides the reader with an overview of MindMeal. It defines the primary operational aspects of the product from the perspectives of end users, maintainers, and administrators. Key features and functions of the system are outlined, along with essential user interactions and interfaces.

2.1 FEATURES & FUNCTIONS

MindMeal is a comprehensive mobile application that integrates intelligent meal management, AI-powered nutritional guidance, and ADHD-specific behavioral support systems. It functions as a centralized hub for eating-related activities, offering personalized recommendations and automated reminders.

Primary Functions:

- **Intelligent Meal Reminders:** A notification system that adapts to user behavior patterns.
- **AI Recipe Generator:** An AI-powered recommendation engine that suggests meals based on dietary restrictions, available ingredients, and cooking skill level.
- **Visual Meal Planning:** A User- friendly interface that categorizes meals and includes preparation instructions.
- **Automated Grocery Management:** Generates a shopping based on previous meals.
- **Grocery Scanner:** Allows users to scan grocery products using their device camera to add items to the inventory, including basic nutritional and product information.

Limitations:

- **Not a Medical Tool:** The application is a support tool and does not replace professional medical advice.
- **Meal Preparation Not Included:** Users are responsible for preparing and consuming meals themselves.
- **Food Logging:** The app depends on the user's input to track meal completion and intake.

System Architecture

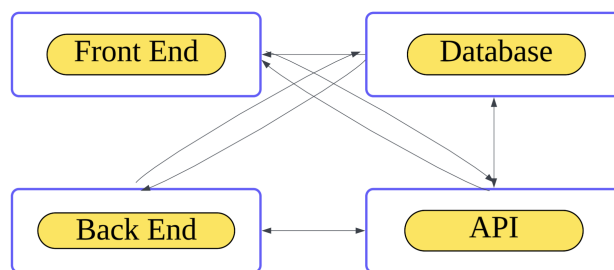


Figure 2: Architecture Diagram

Application Components

- **Mobile Application:** Mobile app for IOS and Android platforms.
- **AI Engine:** AI system for personalized meal recipes and reminders.

- **Notification Management System:** Sends intelligent reminder alert system.
- **Meal Section:** Organized list of suggested meals based on user profile and AI insights.
- **User Profile Management:** Storage of personal data, preferences, and settings.
- **Grocery Shopping List:** Maintains a dynamic list of ingredients needed, based on selected meals and scanned products.

External System Dependencies:

- **Recipe Content APIs:** External services providing access to nutritional data and recipe content.
- **Push Notifications Services:** Firebase Cloud Messaging for delivering real-time reminders.
- **Authentication Services:** Firebase Authentication for secure login and account management.

2.2 EXTERNAL INPUTS & OUTPUTS

This section describes the key inputs and outputs to and from the system.

Name	Description	Type	Source
User Profile Data	Personal information, preferences, and dietary restrictions	Input	Provided by user
Grocery Scanner	Scanned grocery products with barcode and basic info	Input	Device camera
Recipe Preferences	Ratings and feedback on meals	Input	User interaction /
Recipe Recommendations	Personalized meal suggestions with instructions	Output	AI System
Notifications	Meal reminders and behavioral prompts	Output	Application system

Table 2: External Inputs and Outputs

2.3 PRODUCT INTERFACES

This section describes the visible interfaces available to different stakeholders of the system:

- **End User:** The mobile application provides dashboard views for upcoming meals, reminders, and grocery lists. The user interacts through touch gestures, checkboxes, sliders, and intuitive icons.
- **Administrators:** Admins manage user feedback, monitor performance analytics, and oversee API integrations.
- **Maintainers:** System maintainers work with back-end tools to manage database, monitor API, analyze logs, and ensure system stability.

Interface Design Principles

ADHD-Specific Considerations

- **Reduced Cognitive Load:** Limit user interactions to a maximum of 3 to 4 actionable items per screen.
- **Visual Hierarchy:** Use a clear layout structure, size differentiation, and color prioritization to guide the user's attention.
- **Immediate Feedback:** All interactions provide instant visual or haptic feedback to confirm user actions.
- **Consistent Navigation:** Maintain predictable layouts and navigation patterns across all screens to reduce confusion.

2.3.1 DASHBOARD INTERFACE

The dashboard acts as the central hub for user interaction and daily activity tracking.

Interface Specifications:

- High-contrast color scheme to aid in visual focus.
- Large, clearly labeled touch targets for ease of interaction.
- Minimal design to reduce distractions and decision fatigue.
- Consistent iconography paired with text labels for clarity.

2.3.2 MEAL RECOMMENDATION

This screen provides the core functionality for personalized meal suggestions and AI-driven recipe browsing.

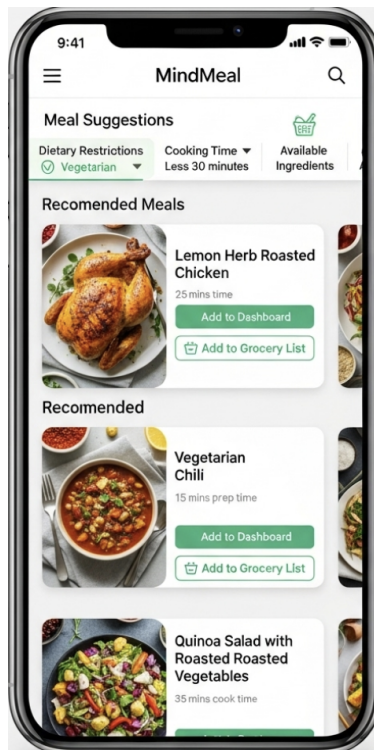


Figure 3: Example of Meal recommendations

Interface Specifications:

- Scrollable list of recommended meals with high-quality images.
- Filters for dietary restrictions, cooking time, and availability of ingredients.
- Option to add meals directly to the dashboard or grocery list.

2.3.3 NOTIFICATIONS INTERFACE

The notifications interface plays a critical role in supporting consistent eating habits by providing timely, context-aware reminders to users.

Interface Specifications:

- Each notification includes a clear action to reduce decision friction.
- Reminders include visual (pop-ups with icons), auditory (notification sounds) , and optional haptic feedback (vibrations) for maximum engagement.
- Undismissed meal reminders appear in a persistent banner at the top of the dashboard until acknowledged or completed.
- Reminders are color-coded by urgency and type (e.g., orange for missed meals, green for upcoming ones).

3 CUSTOMER REQUIREMENTS

This section establishes the specific features and functions required by individuals with ADHD, their caregivers, and healthcare providers who will use MindMeal. Customer requirements define the directly observable "look and feel" and functionality that users will encounter, ensuring the application meets the unique needs of the ADHD community. Each requirement addresses specific challenges faced by people with ADHD in maintaining consistent eating habits, managing executive dysfunction, and coordinating nutrition with medication schedules. Requirements in this section have been developed through consultation with ADHD advocacy groups, healthcare professionals, and potential users, and may only be modified with explicit stakeholder agreement.

3.1 PERSONALIZED MEAL SUGGESTIONS

3.1.1 DESCRIPTION

The application must offer AI-powered meal suggestions that adapt to the user's current food inventory, dietary restrictions, cooking skill level, and available time. These recommendations should appear as visually distinct cards with high quality images, short prep instructions, and clear indicators for ingredients already in the users fridge. The interface must allow the user to filter results based on preferences (e.g., vegetarian, quick meals) and add selected meals to a personal meal plan or grocery list.

3.1.2 SOURCE

User interviews, ADHD advocacy feedback, and team consensus

3.1.3 CONSTRAINTS

The AI system relies on accurate user-provided inventory data. Response time must be kept under 3 seconds for suggestions to feel responsive. Internet connection is required for recipe retrieval. Users with low literacy or executive dysfunction must be able to navigate without reading long blocks of text.

3.1.4 STANDARDS

- Web Content Accessibility Guidelines (WCAG 2.1 – Level AA) for color contrast and text labels
- IEEE 830-1998 for software requirement formatting
- Apple HIG and Material Design Accessibility standards.

3.1.5 PRIORITY

This is one of the core value propositions of the app and defines its intelligent behavior.

3.2 VISUAL FOOD INVENTORY WITH PHOTO AND BARCODE SUPPORT

3.2.1 DESCRIPTION

The system shall provide users with the ability to create and maintain a visual inventory of food items using either barcode scanning or image capture. Each item added must appear in a grid-style visual list that includes a thumbnail image, item name, expiration date, and optional notes (e.g., quantity, storage location). This interface is designed to combat the object permanence issues common in ADHD by making food 'mentally visible.' Inventory should allow filtering by expiration date, category (e.g., dairy, produce), or recently added items, and must be scrollable with large touch targets and readable text labels.

3.2.2 SOURCE

ADHD community surveys, UI testing feedback, Sprint 2 planning discussions.

3.2.3 CONSTRAINTS

Barcode functionality requires third-party APIs for product lookup and may require caching due to rate limits. Image uploads depend on camera access and internet speed, with all assets stored in Firebase Storage. The app must avoid overwhelming visuals, so limits on the number of items shown per screen may apply.

3.2.4 STANDARDS

- WAI-ARIA roles for image labeling
- Google and Apple mobile UX accessibility recommendations
- Google and Apple mobile UX accessibility recommendations
- Firebase CameraX (Android) and AVFoundation (iOS) best practices

3.2.5 PRIORITY

High – This feature addresses a core ADHD-related problem and supports habit formation.

3.3 ADAPTIVE MEAL TASK REMINDER SYSTEM

3.3.1 DESCRIPTION

MindMeal shall include a highly configurable and ADHD-friendly reminder system that prompts users about meals, food expiration, and task nudges (e.g., "Check fridge", "Drink water"). Reminders must support multiple modes: text popups, icon-based cards, optional audio tones, and haptic vibration. Users can choose the time, frequency, and style of reminders, and the app should learn from dismissals or completions to adjust timing accordingly. Missed reminders must persist as banners on the dashboard until addressed, with different colors used to indicate urgency and context (e.g., green for meal due soon, orange for expired food).

3.3.2 SOURCE

Sprint 1 concept presentation, ADHD UX research studies

3.3.3 CONSTRAINTS

Requires user permission to access background notifications, and implementation varies slightly between Android and iOS platforms. Power-saving modes or Do Not Disturb settings may block notifications. Firebase Cloud Messaging will be used, and AI-powered adjustment logic will be limited in MVP version.

3.3.4 STANDARDS

- Firebase Cloud Messaging documentation
- Apple Human Interface Guidelines for notifications
- Google Material Design adaptive UX guidelines
- WCAG 2.1 for sensory feedback and user control

3.3.5 PRIORITY

Critical – This is central to helping users develop routines and remember key tasks.

3.4 SIMPLIFIED, DISTRACTION-FREE DASHBOARD INTERFACE

3.4.1 DESCRIPTION

The system shall present a central dashboard interface that provides a clean, at-a-glance overview of the day's schedule: upcoming meals, active reminders, and recently logged inventory items. This interface must be free from clutter and optimized for users who struggle with decision-making and visual overload. Icons should be large, color-coded and paired with minimal text. There should be no more than 3-4 main options visible at a time (e.g., "View Meals," "Add Food," "Reminders"), and users must be able to reach the dashboard from any screen with one tap. Feedback animations (e.g., "nudge" pulse on expiring items) must be subtle and non distracting.

3.4.2 SOURCE

ADHD accessibility design references, team wireframes

3.4.3 CONSTRAINTS

All dashboard elements must render in under 100ms to avoid UI lag. Layout must adjust across screen sizes (small phones and tablets). Color contrast must follow accessibility standards, and visual notifications must have alternate text or haptics.

3.4.4 STANDARDS

- Apple HIG and Material Design: minimal UI best practices
- W3C guidelines for visual hierarchy and user focus

3.4.5 PRIORITY

High – This is the user's main workspace and must remain simple and supportive.

4 PACKAGING REQUIREMENTS

This section defines how MindMeal will be packaged, distributed, and delivered to end users across different platforms and deployment scenarios. Packaging requirements specify the technical delivery mechanisms, installation processes, branding elements, and distribution channels that will be used to make MindMeal available to individuals with ADHD, their caregivers, and healthcare providers. These requirements ensure consistent user experience across all distribution methods while accommodating the diverse technical capabilities and preferences of the ADHD community. The packaging approach prioritizes accessibility, ease of installation, and immediate usability to minimize barriers for users who may struggle with complex technical procedures.

4.1 MOBILE APPLICATION STORE

4.1.1 DESCRIPTION

MindMeal shall be distributed primarily through official mobile application stores as native iOS and Android applications. The packaging must include:

IOS App Store Distribution:

- Single .ipa package containing the complete MindMeal application
- Universal binary supporting iPhone (iOS 14.0+) and iPad (iPadOS 14.0+)
- Application size optimized to remain under 200MB for initial download
- App Store metadata including ADHD-focused keywords, screenshots, and description
- Privacy nutrition label clearly documenting all data collection practices
- Age rating appropriate for 12+ with parental guidance recommendations

Google Play Store Distribution:

- Android App Bundle (.aab) format for optimized delivery.
- Support for Android 8.0 (API level 26) and higher.
- Adaptive icon design according to Android guidelines.
- Play Console metadata with health and medical category classification.
- Data safety section detailing health information handling.
- Age rating appropriate for 12+ with parental guidance recommendations

4.1.2 SOURCE

Apple App Store Review Guidelines, Google Play Developer Policy Center, Mobile app accessibility best practices, ADHD advocacy group feedback on app accessibility

4.1.3 CONSTRAINTS

- Platform Constraints: Must comply with Apple and Google content policies regarding health applications. Cannot make medical device claims.
- Technical Constraints: App store size limitations require careful asset optimization. Review process can take 1-7 days for updates.
- Economic Constraints: App store fees (30% of premium subscriptions) impact revenue model. Developer account costs (\$99/year iOS, \$25 one-time Android).
- Accessibility Constraints: Must meet platform-specific accessibility requirements for users with motor, visual, or cognitive differences.
- Geographic Constraints: Store availability varies by country, particularly for health-focused applications.

4.1.4 STANDARDS

- Apple App Store Review Guidelines, especially sections on Health Medical apps, Privacy, and Accessibility
- Google Play Developer Policies, specifically policies related to user data, app content, and health-related claims
- WCAG 2.1 guidelines for cognitive and visual accessibility
- ISO/IEC 27001 for secure user data handling practices during installation and updates
- App Store Optimization (ASO) standards for improving visibility and engagement through metadata, keyword tagging, and appropriate content classification

4.1.5 PRIORITY

High —Successful deployment through official app stores is essential for accessibility and widespread adoption of the product. Without proper packaging, the target users (individuals with ADHD) may face barriers to installation and usage, undermining the app's core value proposition.

5 PERFORMANCE REQUIREMENTS

This section defines the specific performance criteria and response time requirements that MindMeal must meet to effectively serve users with ADHD. Performance requirements establish measurable benchmarks for system responsiveness, ensuring that the application delivers smooth, timely interactions across all functional layers. These requirements are critical for maintaining user engagement and minimizing cognitive disruption, as individuals with ADHD are particularly sensitive to delays and system lag that can lead to frustration and app abandonment. Performance metrics have been derived from ADHD-focused usability studies, platform best practices, and international quality standards to ensure optimal user experience across diverse mobile devices and network conditions.

5.1 REAL-TIME INTERACTIVITY AND SYSTEM RESPONSIVENESS

5.1.1 DESCRIPTION

The MindMeal app must deliver smooth, timely interactions across all layers—UI, Data, and Application—to support users with ADHD who are sensitive to lag and frustration from delays. Performance across modules (e.g., inventory scanning, recipe generation, and notification rendering) must remain within strict response thresholds to minimize cognitive disruption.

Key performance metrics include:

- Login & Onboarding: ≤ 3 seconds from action to transition
- Inventory CRUD Operations: ≤ 2 seconds for create/update/delete item
- Recipe Suggestion via LLM: ≤ 3 seconds to display result
- Notifications & Alerts: ≤ 1 second from trigger to device display
- Camera Module (future): ≤ 5 seconds for barcode or image scan processing
- Chat Page Message Rendering: ≤ 200 ms message display latency
- Offline Mode: Core functions like inventory access and saved recipes must remain functional
- Cloud Sync: ≤ 8 seconds for bi-directional synchronization across devices

5.1.2 SOURCE

Derived from usability testing of ADHD-focused applications and requirements in Sections 3.1, 3.2, and 3.3. Also informed by platform guidelines and ISO/IEC 25010 quality models.

5.1.3 CONSTRAINTS

Mobile devices vary widely in memory, CPU, and camera quality. LLM requests rely on API speed and network latency. Offline caching must not exceed 300MB of device storage. Animations must be lightweight and cancelable.

5.1.4 STANDARDS

- iOS and Android performance best practices
- WCAG 2.1 compliance for time-sensitive interactions
- ISO/IEC 25023 performance efficiency metrics
- Firebase Performance Monitoring thresholds

5.1.5 PRIORITY

HIGH â Lag impacts ADHD user trust, satisfaction, and app abandonment.

6 SAFETY REQUIREMENTS

This section defines the critical safety measures and protective mechanisms that MindMeal must implement to ensure user health, well-being, and data security. Safety requirements establish mandatory safeguards for individuals with ADHD and other neurodivergent conditions who may be particularly vulnerable to harmful dietary recommendations, overwhelming notifications, or data loss. These requirements address both immediate safety concerns related to health and nutrition as well as long-term behavioral safety considerations specific to mobile device usage patterns among neurodivergent users. All safety measures must prioritize user protection while maintaining the application's core functionality and accessibility features, ensuring that MindMeal serves as a supportive tool without introducing additional risks or stressors into users' daily routines.

6.1 USER HEALTH & NUTRITIONAL SAFETY COMPLIANCE

6.1.1 DESCRIPTION

The app must ensure dietary suggestions and reminders do not endanger user health. The recommendation engine must avoid allergens listed in user-provided data, and display clear warnings in recipe outputs if allergen risks or conflicting medical conditions are detected. All recommendations must include disclaimers that the app does not replace professional medical advice.

6.1.2 SOURCE

Derived from features in Section 2.1.2 (Personal Info + Recipe Maker), and the application's intended audience involving medically sensitive individuals.

6.1.3 CONSTRAINTS

App does not diagnose or treat medical conditions. Data entry inaccuracies may impact safety. Internet dependency for real-time allergy rule enforcement.

6.1.4 STANDARDS

- FDA mobile health app guidelines
- HIPAA data usage restrictions
- WCAG cognitive safety and UI fallback provisions

6.1.5 PRIORITY

HIGH – Legal and ethical obligation to prevent harmful recommendations.

6.2 MOBILE DEVICE AND NOTIFICATION SAFETY

6.2.1 DESCRIPTION

MindMeal must ensure that its notification, alert, and reminder systems do not interfere with safe mobile device usage or user well-being. This includes preventing distractions while driving or sleeping and minimizing potential over-reliance on phone-based cues that may induce stress or anxiety in neurodivergent users. Additionally, repeated notifications must not create compulsive interaction patterns or contribute to screen fatigue.

Notifications should be gentle, adjustable, and fully opt-in, with user-customizable intensity and quiet hour settings. The system should avoid vibration-only alerts for users with sensory processing issues, and provide calming tone options where possible.

6.2.2 SOURCE

Derived from ADHD user feedback on digital overload and cognitive stress related to excessive notifications. Supported by clinical research on mobile device distraction risks and mental health guidelines for digital therapeutic tools.

6.2.3 CONSTRAINTS

Must comply with Android and iOS Do Not Disturb APIs. Alerts must respect scheduled quiet periods (e.g., sleep, focus time). Users must be able to mute or disable all alerts per category. Emergency alerts (if any) must be clearly labeled and justified. Push frequency must be rate-limited per user preferences.

6.2.4 STANDARDS

- Apple Human Interface Guidelines: Notifications
- Android Developer Documentation: NotificationManager
- WHO guidelines for digital health intervention safety
- IEEE 1474.1 standards on alert fatigue and notification prioritization

6.2.5 PRIORITY

MEDIUM – Important for sustained user comfort and behavioral safety, especially for neurodivergent users.

6.3 SYSTEM INTEGRITY AND DATA RECOVERY SAFETY

6.3.1 DESCRIPTION

The MindMeal application must protect users from data loss, corrupted files, or unintended behavior caused by device failure, app crashes, or interrupted synchronization processes. Users with ADHD may struggle to recall past entries or repeated setup steps, so preserving critical information (e.g., personal info, inventory, past recipes) is essential.

The app must include failsafes for local caching, encrypted backups, and auto-recovery features. Inventory and health data must not be lost due to app uninstalls, version upgrades, or sync interruptions. Any data rollback or migration should provide clear prompts and confirmations.

6.3.2 SOURCE

This requirement stems from cognitive load considerations and usability studies with ADHD users, where re-entering lost data significantly reduces continued app use. Technical resilience is vital for health apps handling user-specific, time-sensitive information.

6.3.3 CONSTRAINTS

Offline local data must be recoverable on reinstall (if not explicitly deleted). Sync operations must implement transactional integrity. Conflict resolution strategies must minimize user confusion. Data migration between versions must preserve compatibility.

6.3.4 STANDARDS

- OWASP Mobile Security Testing Guide (MSTG-STORAGE and MSTG-RESILIENCE)
- ISO/IEC 27001 for data integrity and availability
- NIST SP 800-34 for contingency planning
- Android Room / iOS CoreData resilience practices

6.3.5 PRIORITY

HIGH – Loss of critical user data is unacceptable in healthcare-oriented apps, particularly for neurodiverse users who rely on consistent system behavior.

7 MAINTENANCE & SUPPORT REQUIREMENTS

Following the initial deployment of the MindMeal mobile application, long-term support and maintenance will be necessary to ensure continued usability, stability, and security. Because the app targets neurodivergent users, any disruption in core functionality (e.g., reminders, inventory logging) could significantly affect user experience. The project team must ensure that technical documentation is preserved, the codebase remains accessible and maintainable, and future developers can troubleshoot issues or enhance the system without extensive rework. Maintenance includes addressing bug fixes, updating libraries (especially for ReactNative and Firebase), managing backend service dependencies (Python, APIs), and incorporating accessibility updates based on user feedback. All documentation, deployment scripts, and test environments must be version-controlled and accompanied by detailed developer guidance to enable smooth handoff and ongoing support.

7.1 SOURCE CODE DOCUMENTATION AND AVAILABILITY

7.1.1 DESCRIPTION

The source code for MindMeal must be maintained in a version-controlled GitHub repository. The repository should include detailed comments, inline documentation, and a developer README to support onboarding and future troubleshooting. The codebase shall be modular and follow industry-standard naming conventions and file structures to improve maintainability. The repository must be open to project stakeholders with permissions managed through GitHub roles. All code must follow a documented commit policy, and each module must be independently

7.1.2 SOURCE

CSE Senior Design Course Requirements, GitHub Best Practices, team planning discussion

7.1.3 CONSTRAINTS

Contributors must follow the existing documentation and architecture pattern. Updates may require local testing before merging. Hosting and GitHub Actions must remain within free-tier limitations. Compatibility with development tools, frameworks and environments must be maintained for at least 12 months.

7.1.4 STANDARDS

- IEEE 14764:2006 for Software Maintenance
- GitHub Flow workflow and commit policies
- Markdown-based documentation for compatibility

7.1.5 PRIORITY

High – Proper maintenance of source code is essential to long-term success, future improvements, and knowledge transfer.

8 OTHER REQUIREMENTS

This section defines additional requirements that are essential to MindMeal's success but do not fall into the previous categories of customer, packaging, performance, safety, or maintenance requirements. These requirements encompass cross-cutting concerns that affect the entire application ecosystem, including accessibility standards, compliance mandates, and technical specifications that enable the app to serve its target demographic effectively. Other requirements establish the foundational capabilities that make MindMeal truly accessible and usable for individuals with ADHD and co-occurring neurodivergent conditions, ensuring that the application not only meets functional needs but also provides an inclusive and supportive user experience that accommodates diverse cognitive and sensory processing differences.

8.1 ACCESSIBILITY FOR NEURODIVERSE USERS

8.1.1 DESCRIPTION

MindMeal must be fully accessible to users with ADHD and potentially co-occurring conditions like dyslexia, sensory sensitivity, and executive dysfunction. Interfaces must minimize distractions and follow cognitive accessibility heuristics. App must support screen readers, one-handed use, voice input (where feasible), and text resizing.

Design features must include:

- Reduced animation and toggles to disable them
- Large, well-spaced touch targets ($\geq 44\text{px}$)
- Contextual tooltips and onboarding guidance
- Dark mode and high contrast modes
- Optional visual simplification toggle (Minimalist Mode)

8.1.2 SOURCE

Drawn from Sections 2.3.1 and 3.4.1 and industry guidance (WCAG Cognitive Accessibility Roadmap and Gap Analysis, ADHD design research)

8.1.3 CONSTRAINTS

Accessibility must not interfere with performance. UI/UX decisions must balance minimalism with functionality. Localization efforts must retain accessibility.

8.1.4 STANDARDS

- WCAG 2.1 AA
- ISO/IEC 40500:2012
- ADA guidelines for digital content
- Apple's Accessibility Guidelines / Android Accessibility Suite

8.1.5 PRIORITY

HIGH – Central to app mission and target demographic usability.

9 FUTURE ITEMS

This section outlines features and capabilities that are planned for future releases of MindMeal but are not included in the current development scope or minimum viable product (MVP). Future items represent strategic enhancements that will expand the application's value proposition and ecosystem integration capabilities, particularly in the healthcare and digital therapeutics domains. These requirements have been identified through stakeholder feedback, market research, and emerging trends in digital health solutions for neurodivergent populations. While not immediately essential for the core MindMeal experience, these future items establish a roadmap for long-term product evolution and potential partnerships with healthcare providers, ensuring that the application can grow to meet expanding user needs and regulatory requirements in the digital health landscape.

9.1 HEALTH SYSTEM INTEGRATION & PROVIDER ACCESS

9.1.1 DESCRIPTION

Future versions of MindMeal shall support integration with electronic health records (EHRs) and care provider portals. With user consent, dietary logs, inventory behavior, and reminder adherence data will be securely shareable with doctors, dietitians, or therapists to inform holistic treatment.

Potential functionalities:

- PDF export or secure FHIR-compliant data sharing
- Alert systems for missed meals or risky eating patterns
- Provider feedback loop to suggest dietary interventions

9.1.2 SOURCE

Proposed based on emerging digital therapeutic trends and ADHD patient care requirements. Mentioned by user stakeholders in preliminary road map planning.

9.1.3 CONSTRAINTS

Strict HIPAA and GDPR compliance. User opt-in consent. EHR vendor API availability and compatibility.

9.1.4 STANDARDS

- HL7 FHIR (Fast Healthcare Interoperability Resources)
- OAuth 2.0 / SMART on FHIR authorization
- HITRUST CSF certification
- Regional telehealth and data exchange regulations

9.1.5 PRIORITY

MEDIUM – Strategic road map goal for future value but not required for MVP launch.

REFERENCES